



Beams and structures

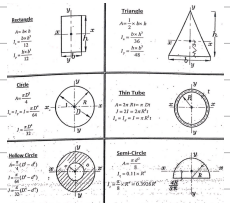
→ How to solve beams questions:

- ① Draw Free body diagram, Find any unknown reaction using the equilibrium equations
- ② Plot the Load Diagrams from the Free body diagram: "NFD, SFD, BMD"
- ③ Determine the critical section or use the section mentioned if given, get the centroid of it
 - Critical section is the section at maximum Bending moment & zero shear force
 - Centroid = $(\bar{x}, \bar{y}) \Rightarrow \bar{x} = \frac{\sum A_i x_i}{\sum A_i}$, $\bar{y} = \frac{\sum A_i y_i}{\sum A_i}$
- ④ identify the Loading Condition we have to calculate the area Property corresponding to it :

$V_x \rightarrow I_y, Q$
 $V_y \rightarrow I_x, Q$
 $I_x = \sum [I_{x_i} + A(x_i^2 + y_i^2)]$

$F \rightarrow A$
 $M_x \rightarrow I_x$
 $I_y = \sum [I_{y_i} + A(y_i^2 + x_i^2)]$

$M_y \rightarrow I_y$
 $T \rightarrow J, Q$
 $J = I_x + I_y$



- ⑤ Calculate the normal stress and shear stress

$$\tau = \frac{VQ}{It} + \frac{T_r}{J}, \quad \sigma = \pm \frac{F}{A} \pm \frac{M_x}{I_x} y \pm \frac{M_y}{I_y} x$$

- ⑥ Plot the stress diagrams "NSD, SSD, TD"

- ⑦ Draw the inferential elements

→ How to solve the structures problems:

- ① Determine the loading conditions: you need to specify the three axis and you may need to analyse the internal forces to get the loading conditions
- ② Calculate the area Property corresponding to the determined loading conditions
- ③ Calculate the normal stress and shear stress

if the cross section is circular we can consider $\sigma = \frac{F}{A} \pm \frac{M_r}{I} c$, where $I = I_x = I_y$, $M_r = \sqrt{M_x^2 + M_y^2}$
- ④ Plot the stress diagrams
- ⑤ Draw the inferential elements

Principal stresses & inferential elements

* Steps to solve problems:

① Get $\sigma_x, \sigma_y, \tau_{xy}, \theta \rightarrow$ "if given"

→ if θ is given:

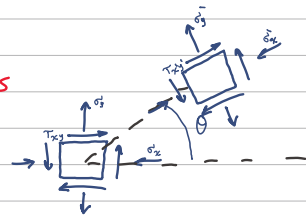
② Calculate $\sigma_x', \sigma_y', \tau_{xy}'$:

$$\sigma_x' = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta + \tau_{xy} \sin 2\theta \quad \sigma_y' = \frac{\sigma_x + \sigma_y}{2} - \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$\tau_{xy}' = -\frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \cos 2\theta$$

③ Plot the inferential element using the determined values

→ if θ is not given:



② Calculate $\sigma_{max}, \sigma_{min}, \theta_p$:

$$\sigma_{max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \sigma_{min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

③ Substitute θ_p in σ_x' then compare the result with $\sigma_{max}, \sigma_{min}$ you will find it equals to one of them, so σ_y' will be equal to the other

→ ex: if $\sigma_{max} = -25 \text{ MPa}, \sigma_{min} = 12 \text{ MPa}, \sigma_x' = 12 \text{ MPa}$, then $\sigma_y' = -25 \text{ MPa}$

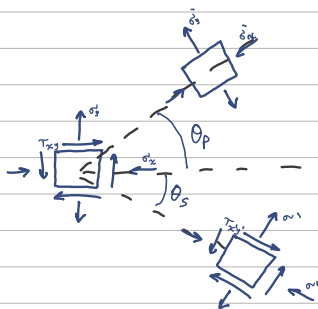
④ Calculate τ_{max}, τ_{min} :

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \quad \tau_{min} = -\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

⑤ Calculate $\theta_s, \tau_{xy}', \sigma'$:

$$\tan 2\theta_s = -\frac{\sigma_x - \sigma_y}{2\tau_{xy}} \quad \sigma' = \sigma_{av} = \frac{\sigma_x + \sigma_y}{2}$$

⑥ Plot 2 inferential elements:



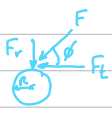
shafts

* Steps to solve shaft problems:

① Find torque and plot the torque diagram:

→ $T = \frac{P}{\omega}$ → having Power, ω

→ $T = \frac{9550 \times P(\text{kW})}{N(\text{rpm})}$ → having Power, rpm

→ $T = F_t \times R$ → having a gear :  , $\tan \phi = \frac{F_r}{F_t}$

→ $T = F \times R$ → having a sprocket drive : 

→ $T = (F_1 - F_2) R$ → having a pulley drive : $\frac{F_1}{F_2} = e^{\mu \theta}$  $\theta = 180 - 2 \sin^{-1} \left(\frac{D-d}{2c} \right)$
 $c = \text{distance between two pulleys}$

② Plot vertical & horizontal planes equilibrium diagrams

③ Plot M_y diagram from horizontal plane & M_x diagram from vertical plane

④ Determine the critical section having the highest T , M_r : $M_r = \sqrt{M_x^2 + M_y^2}$

⑤ Calculate the shaft diameter according to the specified failure theory mentioned in the problem

→ Maximum principal normal stress theory

$$\sigma_{\max} \leq \sigma_{\text{all}} \Rightarrow \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \leq \frac{\sigma_y}{n} \Rightarrow \left\{ \frac{16}{\pi d^3} \left[K_b M + \sqrt{(K_b M)^2 + (K_t T)^2} \right] \right\}$$

used only if F_{50} and cross section is solid circle

→ Maximum principal shear stress theory

$$\tau_{\max} \leq \tau_{\text{all}} \Rightarrow \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \leq \frac{\sigma_y}{2n} \Rightarrow \left\{ \frac{16}{\pi d^3} \sqrt{(K_b M)^2 + (K_t T)^2} \right\}$$

used only if F_{50} and cross section is solid circle

→ Maximum principal strain theory : $\sigma_{\max} - \nu \sigma_{\min} \leq \frac{\sigma_y}{n}$

→ Maximum principal strain energy theory : $\sigma_{\max}^2 + \sigma_{\min}^2 - 2\nu \sigma_{\max} \sigma_{\min} \leq \left(\frac{\sigma_y}{n}\right)^2$
 $\hookrightarrow \approx 0.3$

⑥ Calculate angle of twist (θ) : $\theta = \sum \frac{TL}{JG}$

$T \Rightarrow$ torque , $L \Rightarrow$ length affected by the torque

$J \Rightarrow I_x + I_y$, $G \Rightarrow$ modulus of elasticity (80 GPa)